Interpreting geometric language

Teacher resource

Question 1

Daisy made a skeletal model of a square based pyramid. She used 8 plastic straws of equal length and 5 joiners to hold the straws at the vertices of the pyramid.

She wishes to change this model to make a skeletal model of a cube.

How many more straws and joiners will she need?

- ⬭ 3 straws and 4 joiners
- ⬭ 4 straws and 3 joiners
- ⬭ 5 straws and 8 joiners
- ⬭ 12 straws and 8 joiners

Answer: B
Skill: Students visualise familiar 3D objects given a description using geometric language.
Additional questions

1. What shaped pyramid can be made with 16 straws and 9 joiners?
2. How many more straws and joiners are needed to change this pyramid into an octagonal prism?

Question 2

Three angles meet at a point. The sum of the three angles is 200°.

Which of the following statements are true?

- ⬭ The three angles could all be obtuse
- ⬭ One of the angles could be a reflex angle
- ⬭ The three angles could all be acute
- ⬭ One of the angles could be a straight angle

Answer: B, C and D
Skill: Students interpret geometric language.
Background information
Some secondary students may be able to interpret geometric language and symbols when used by others but not be able to use either independently. Students may benefit from experiences of using specific geometric language for themselves and constructing diagrams and accurate geometric drawings which incorporate conventional geometric symbols. Students are supported to develop their understanding of concepts as they use the language and symbols required to describe those concepts.

Additional questions

1. Which statements would be true if the sum of the three angles was 280°?
2. Which statements would be true if the sum of the three angles was 380°?

Question 3

Which of the following figures have perpendicular diagonals?

- ⬭ rectangles
- ⬭ squares
- ⬭ kites
- ⬭ trapeziums

Answer: B and C
Skill: Students interpret geometric language

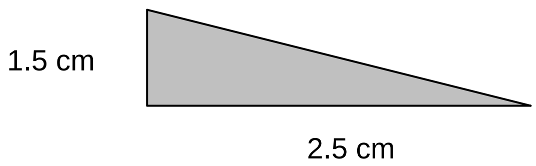
Additional questions

1. Which of the figures have diagonals that bisect each other?
2. Which of the figures have at least one pair of parallel sides?

Question 4

A jeweller cuts sheets of silver into right angled triangles to make necklaces

The dimensions of the shorter sides of these triangles are shown in the diagram below.



What is the maximum number of these triangles she can cut from a rectangular sheet of silver which is 20 cm long and 12 cm wide, without having any silver left over?

Answer: 128
Skill: Students interpret geometric language in a context.

Additional questions

1. Daisy could also cut the sheet into identical right angled isosceles triangles. What is the length of the short sides of the largest right angled isosceles triangles she could cut without having any silver left over?
2. What is the maximum number of these right angled isosceles triangles that she could cut from the sheet?

Student activity: Interpreting geometric language

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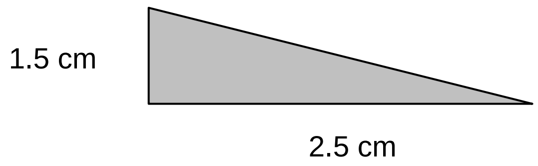

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